CONTROL SYSTEM LAB-7

BLOCK DIAGRAM REDUCTION

SOWRAV CHOWDHURY

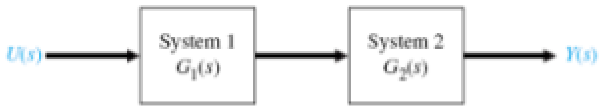
183016008

I. Experiment Title: Block Diagram Reduction

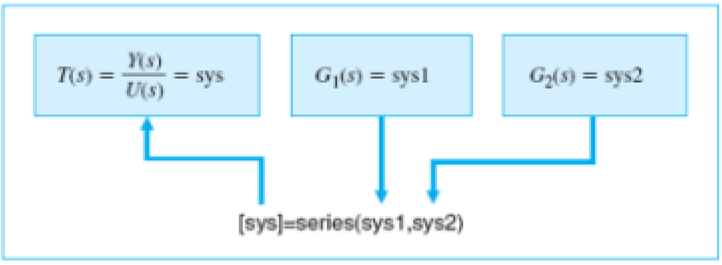
II. Objectives: The objective of this exercise will be to learn commands in MATLAB that would be used to reduce linear systems block diagram using series, parallel and feedback configuration.

III. Theory:

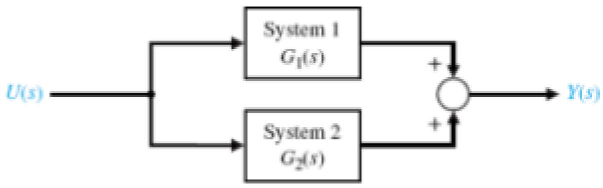
Series configuration: If the two blocks are connected as shown below then the blocks are said to be in series. It would like multiplying two transfer functions. The MATLAB command for the such configuration is "series".



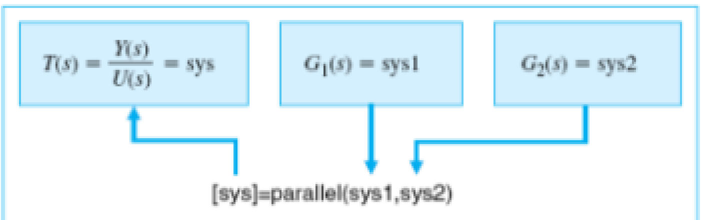
The series command is implemented as shown below:



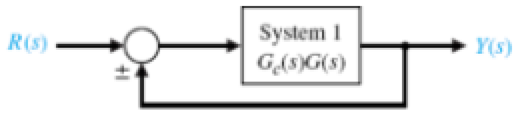
Parallel configuration: If the two blocks are connected as shown below then the blocks are said to be in parallel. It would like adding two transfer functions.



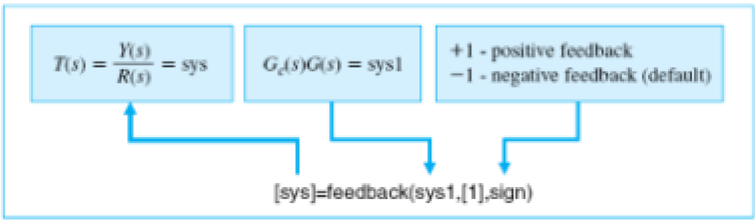
The MATLAB command for implementing a parallel configuration is "parallel" as shown below:



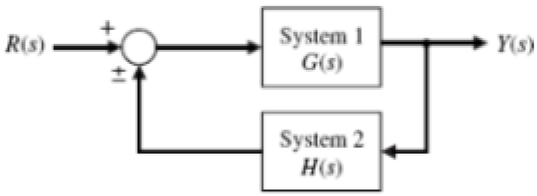
Feedback configuration: If the blocks are connected as shown below then the blocks are said to be in feedback. Notice that in the feedback there is no transfer function H(s) defined. When not specified, H(s) is unity. Such a system is said to be a unity feedback system.



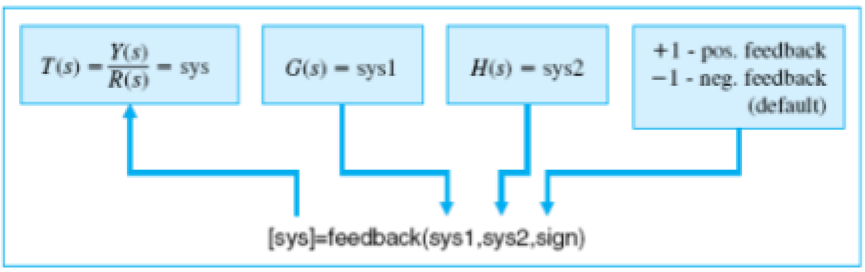
The MATLAB command for implementing a feedback system is "feedback" as shown below:



When H(s) is non-unity or specified, such a system is said to be a non-unity feedback system as shown below:

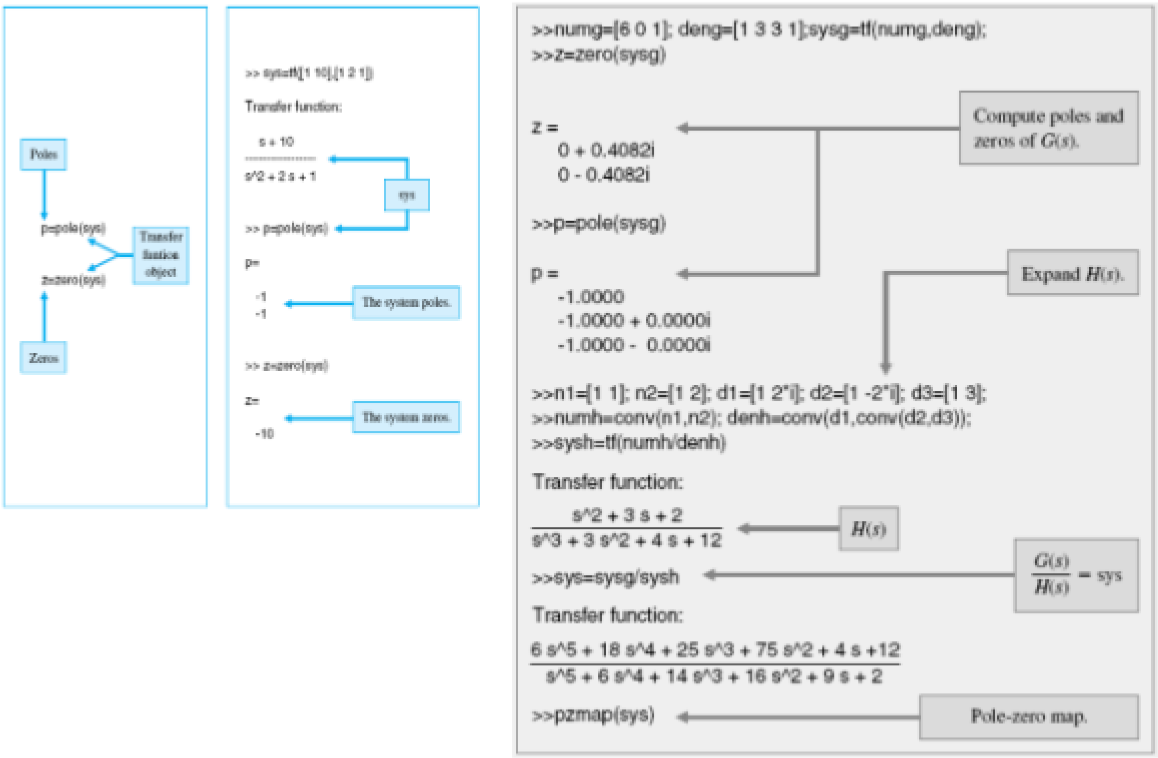


A non-unity feedback system is implemented in MATLAB using the same "feedback" command as shown:



Poles and Zeros of System: To obtain the poles and zeros of the system use the MATLAB command "pole" and "zero" respectively as shown in example 5. You can also use MATLAB command "pzmap" to obtain the same.

Example 5: Given a system transfer function plot the location of the system zeros and poles using the MATLAB pole-zero map command.



IV. Lab Work:

a. **Example 1:** Given the transfer functions of individual blocks generate the system transfer function of the block combinations.

```
clc; clear all; close all;
num1 = [1 1];
den1 = [1 2];
sys1 = tf(num1, den1)

num2 = [1];
den2 = [500 0 0];
sys2 = tf(num2, den2)
sys = series(sys2,sys1)

sys =
        s + 1
  ------------------
  500    3 + 1000 s^2
```

b. **Example 2:** For the previous systems defined, modify the MATLAB commands to obtain the overall transfer function when the two blocks are in parallel.

```
clc; clear all; close all;
num1 = [1 1];
den1 = [1 2];
sys1 = tf(num1, den1)

num2 = [1];
den2 = [500 0 0];
sys2 = tf(num2, den2)

sys = parallel(sys2,sys1)

sys =

  500 s^3 + 500 s^2 + s + 2
  -------------------------
     500 s^3 + 1000 s^2
```

c. **Example 3:** Given a unity feedback system as shown in the figure, obtain the overall transfer function using MATLAB:

```
clc; clear all; close all;
num1 = [1 1];
den1 = [1 2];
sys1 = tf(num1, den1)

num2 = [1];
den2 = [500 0 0];
sys2 = tf(num2, den2)

sys3 = series(sys1, sys2);
sys = feedback(sys3, [1])

sys =

            s + 1
  --------------------------
  500 s^3 + 1000 s^2 + s + 1
```

d. **Example 4:** Given a non-unity feedback system as shown in the figure, obtain the overall transfer function using MATLAB:

```
clc; clear all; close all;

num1 = [1];
den1 = [500 0 0];
sys1 = tf(num1, den1)

num2 = [1 1];
den2 = [1 2];
sys2 = tf(num2, den2)

sys = feedback(sys1, sys2)
```

```
sys =

             s + 2
  --------------------------
  500 s^3 + 1000 s^2 + s + 1
```

e. **Example 5:** Given a system transfer function plot the location of the system zeros and poles using the MATLAB pole-zero map command.

```
clc; clear all; close all;
num1 = [6 0 1];
den1 = [1 3 3 1];
sys1 = tf(num1, den1);
z = zero(sys1)
p = pole(sys1)
```

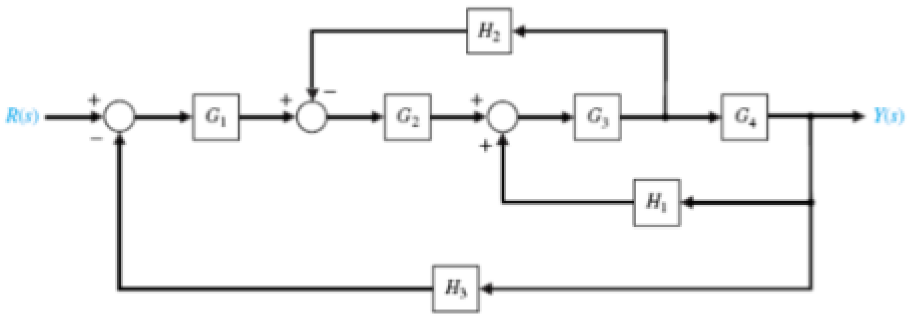
```
z =

   0.0000 + 0.4082i
   0.0000 - 0.4082i


p =

  -1.0000 + 0.0000i
  -1.0000 - 0.0000i
  -1.0000 + 0.0000i
```

Exercise 1: For the following multi-loop feedback system, get closed loop transfer function and the corresponding pole-zero map of the system.



Given $G_1 = \frac{1}{(s+10)}$; $G_2 = \frac{1}{(s+1)}$; $G_3 = \frac{s^2+1}{(s^2+4s+4)}$; $G_4 = \frac{s+1}{(s+6)}$; $H_1 = \frac{s+1}{(s+2)}$; $H_2 = 2$

; $H_3 = 1$ (

MATLAB CODE:

```
clc; clear all; close all;

numg1 = [1];
deng1 = [1 10];
sysg1 = tf(numg1,deng1)

numg2 = [1];
deng2 = [1 1];
sysg2 = tf(numg2,deng2)

numg3 = [1 0 1];
deng3 = [1 4 4];
sysg3 = tf(numg3,deng3)

numg4 = [1 1];
deng4 = [1 6];
sysg4 = tf(numg4,deng4)

numh1 = [1 1];
denh1 = [1 2];
sysh1 = tf(numh1,denh1)

numh2 = [2];
denh2 = [1];
sysh2 = tf(numh2,denh2)

numh3 = [1];
denh3 = [1];
sysh3 = tf(numh3,denh3)
```

```
sys2 = series(sysg3, sysg4);

sys3 = feedback(sys2, sysh1, +1);
sys4 = series(sysg2, sys3);
sys1 = sys2/sys4;
sys5 = feedback(sys4, sys1);
sys6 = series(sysg1, sys5);
sys = feedback(sys6, [1])


sys =

            s^5 + 4s^4 + 6s^3 + 6s^2 + 5s + 2
  ------------------------------------------------------------
  12s^6 + 205s^5 + 1066s^4 + 2517s^3 + 3128s^2 + 2196s + 712
```

V. Discussion:

The use of **block diagrams** to illustrate a cause-and-effect relationship is prevalent in control. We use operational blocks to represent transfer functions and lines for unidirectional information transmission. It is a nice way to visualize the interrelationships of various components. They will be crucial in helping us identify manipulated and controlled variables and input(s) and output(s) of a system.

Follow these rules for simplifying (reducing) the block diagram, which is having many blocks, summing points and take-off points.

- **Rule 1** − Check for the blocks connected in series and simplify.
- **Rule 2** − Check for the blocks connected in parallel and simplify.
- **Rule 3** − Check for the blocks connected in feedback loop and simplify.
- **Rule 4** − If there is difficulty with take-off point while simplifying, shift it towards right.
- **Rule 5** − If there is difficulty with summing point while simplifying, shift it towards left.
- **Rule 6** − Repeat the above steps till you get the simplified form, i.e., single block.

501**Theory:** The buck–boost converter is a type of DC-to-DC converter that has an output voltage magnitude that is either greater than or less than the input voltage magnitude. It is equivalent to a flyback converter using a single inductor instead of a transformer.[1] Two different topologies are called buck–boost converter. Both of them can produce a range of output voltages, ranging from much larger (in absolute magnitude) than the input voltage, down to almost zero.

In the inverting topology, the output voltage is of the opposite polarity than the input. This is a switched-mode power supply with a similar circuit topology to the boost converter and the buck converter. The output voltage is adjustable based on the duty cycle of the switching transistor. One possible drawback of this converter is that the switch does not have a terminal at ground; this complicates the driving circuitry. However, this drawback is of no consequence if the power supply is isolated from the load circuit (if, for example, the supply is a battery) because the supply and diode polarity can simply be reversed. When they can be reversed, the switch can be on either the ground side or the supply side.

1. Construct the buck-boost converter circuit using SIMULINK and follow the steps of Lab Work 1 · Input voltage = 30 V and Intended output voltage = 10, 20, 30, 50, 100 V

· Frequency = 10 KHz and 25 KHz

· Determine the **duty cycle, R, L, C** values as you design the circuit.

Show the following results using two scopes:

· Source Voltage
· Load Voltage
· Inductor Voltage
· Load(output) Current
· Input Current
· Inductor Current

Comment on the nature of the results for different frequencies and different intended output voltages.

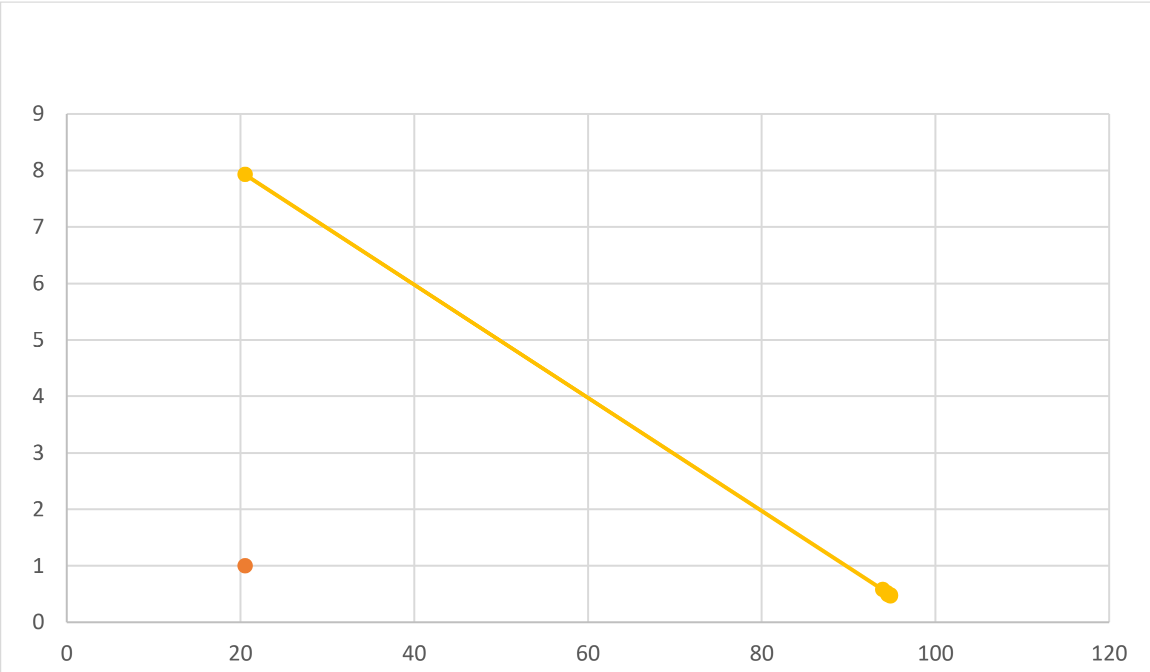
Calculation Table:

Sl No.	Frequency	Input	Output	R	L	C	Duty Cycle
		30	10				
			20				

502Experiment: Construct the circuits as shown and obtain the I-V characteristic of the device as indicated,

1. BJT as a Switch: (The transistor under test is either BD243 or 2N3035 npn transistor)

	V_{BB}	V_{BE}	I_B	I_C	V_{CE}
1	1	0.633	0.2	20.5	7.93
2	2	0.699	1.2	93.9	0.580
3	3	0.713	2.2	94.4	0.523
4	5	0.737	4.1	94.5	0.492
5	8	0.772	7.2	94.8	0.488
6	10	0.792	9.2	94.8	0.467


9
8
7
6
5
4
3
2
1
0
0
20
40
60
80
100
120